

EX. NO: 10

IMPLEMENTATION OF SHA-1 HASHING

AIM

To write a C program to implement the SHA-1 hashing technique.

ALGORITHM

Step 1: Start the program.

Step 2: Read the input message from the user.

Step 3: Convert the message into binary format.

Step 4: Pad the message by:

- Appending a single 1 bit.
- Appending 0 bits until the message length becomes 64 bits less than a multiple of 512 bits.
- Appending the original message length as a 64-bit value.

Step 5: Divide the padded message into 512-bit blocks.

Step 6: Initialize the five 32-bit hash variables (H0, H1, H2, H3, H4) with predefined constants.

Step 7: Process each 512-bit block through 80 rounds of SHA-1 operations using logical functions, rotations, and constants.

Step 8: Combine the final hash values to produce a 160-bit message digest.

Step 9: Display the generated SHA-1 hash value.

Step 10: Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char message[100];
```

```
    unsigned long hash = 0;
```

```
    int i;
```

```
    printf("Enter Message: ");
```

```
    scanf("%s", message);
```

```
// Simple hash (Demonstration only)

for(i = 0; message[i] != '\0'; i++)
{
    hash = hash * 31 + message[i];
}

printf("\nSHA-1 Hash (Demo): %08lX\n", hash);

return 0;
}
```

OUTPUT:

The screenshot displays the Programiz Online C Compiler interface. The browser address bar shows the URL `programiz.com/c-programming/online-compiler/`. The compiler's header includes the Programiz logo, a banner for "LAUNCH INTO THE TECH WORLD" with an image of a man, and the Adelaide University logo. A "Programiz PRO" button is visible in the top right.

The main editor area shows a C program in `main.c` with the following code:

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char message[100];
7     unsigned long hash = 0;
8     int i;
9
10    printf("Enter Message: ");
11    scanf("%s", message);
12
13    // Simple hash (Demonstration only)
14    for(i = 0; message[i] != '\0'; i++)
15    {
16        hash = hash * 31 + message[i];
17    }
18
19    printf("\nSHA-1 Hash (Demo): %08lX\n", hash);
20
21    return 0;
22 }
```

The "Output" panel on the right shows the results of the program execution:

```
Enter Message: pallavi
SHA-1 Hash (Demo): 17D0647E03
=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the system clock indicating 09:28 PM on 03-08-2026, and the weather showing 27°C and "Mostly cloudy".